NFTransUNet: A Neuro-Fuzzy Deep Learning Model for Breast Tumor Segmentation in Big Data Environment

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Abstract. Segmentation of breast tumors from mammograms plays a crucial role in computer-aided diagnosis, enabling precise lesion localization and aiding in early breast cancer detection. However, the diversity in shape, irregular boundaries, and varying sizes of breast tumors remains a significant challenge for existing deep learning models. This paper proposes NFTransUNet, a hybrid deep learning architecture integrating convolutional neural networks, a global contextual Transformer-based model, and a neuro-fuzzy inference module to improve segmentation performance. The proposed model is built upon a fuzzy inference system with Gaussian membership functions, enabling adaptive modeling of complex feature relationships while preserving boundary information. By combining local feature extraction, long-range dependency learning, and fuzzy inference, NFTransUNet enhances the efficiency of representing ambiguous tumor regions. To support large-scale mammography analysis, the proposed model is deployed in a distributed Apache Spark environment, enabling scalable training and inference across multiple computing nodes. Extensive experiments conducted on the public VinDr-Mammo dataset and the internal VL-Bre dataset demonstrate the effectiveness of the proposed approach. NFTransUNet achieved a Dice of 0.935 on the VL-Bre dataset and 0.930 on the VinDr-Mammo dataset, improving performance compared to traditional network models while maintaining efficient distributed processing for medical imaging applications.

Keywords: Breast tumor segmentation · Mammography · Neuro-fuzzy inference · Fuzzy deep learning · Distributed training

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D. T. C. Lai et al. (Eds.): MIWAI 2026, LNAI 17148, pp. 341–352, 2027.
https://doi.org/10.1007/978-981-92-6510-7_26

1 Introduction

According to 2022 global cancer statistics, breast cancer remains one of the three most frequently diagnosed cancers worldwide [2]. Mammography is the primary imaging modality for breast cancer screening, and deep learning has become an essential tool for computer-aided detection and diagnosis [3,15]. Among these tasks, accurate pixel-level tumor segmentation plays a crucial role in lesion localization and subsequent clinical assessment [7]. Encoder-decoder architectures, including U-Net [9], U-Net++ [17], and TransUNet [4], have achieved promising performance in mammographic tumor segmentation. However, breast lesions often exhibit indistinct boundaries and gradual transitions between normal tissue, benign abnormalities, and malignant masses. Conventional binary segmentation masks therefore provide only an implicit representation of this inherent uncertainty, making accurate delineation challenging. Fuzzy logic offers a natural framework for modeling such ambiguity by allowing each pixel to possess a degree of membership rather than a strict binary assignment. Recent studies have demonstrated that incorporating fuzzy logic into deep neural networks can improve segmentation robustness in uncertain and low-contrast imaging scenarios [6,12,13]. Several fuzzy-enhanced segmentation approaches have been reported in the literature. Ma et al. [6] integrated a fuzzy logic unit into the skip connections of U-Net for remote-sensing image segmentation, achieving approximately 92% IoU. Pani and Chawla [8] incorporated an ANFIS module into a Fuzzy Inception U-Net for multimodal brain tumor MRI segmentation and reported a Dice score of 89.66%. In related work [13], fuzzy-logic-based preprocessing was combined with U-Net and Mask R-CNN for mammographic mass segmentation, yielding a Dice score of 92.5% on a multi-hospital dataset. Verma et al. [14] introduced an intuitionistic fuzzy extension of U-Net, whereas Anari et al. [1] combined attention mechanisms with multiple U-Net variants for ultrasound tumor segmentation. Furthermore, the survey by Talpur et al. [12] concluded that integrating fuzzy techniques with deep learning models can improve segmentation accuracy by approximately 11.6% compared with conventional deep networks. Despite these advances, another practical challenge arises from the rapidly growing volume of mammography data. Training deep segmentation models on a single computing node is often time-consuming, motivating the adoption of distributed computing frameworks to accelerate large-scale model training [16].

This paper investigates a neuro-fuzzy layer placed at the encoder bottleneck of a deep segmentation network for breast tumor segmentation under distributed, big-data training. The main contributions of this paper are presented as follows:

- Proposal of NFTransUNet, an efficient mammogram segmentation model that integrates a neuro-fuzzy layer with learnable Gaussian membership functions and rule weights into the encoder bottleneck, enabling enhanced feature representation and lesion delineation (Sect. 3).
- Systematic evaluation of the proposed approach with three backbone architectures (U-Net, U-Net++, and TransUNet) using both the public VinDr-Mammo dataset and the internal VL-Bre dataset (Sec. 4).

– An experimental evaluation of the proposed model in an Apache Spark distributed environment, assessing its speedup, parallel efficiency, and scalability through serial-fraction analysis on a multi-node cluster (Sect. 4.3).

2 Background

2.1 Learning-Based Segmentation

U-Net [9] follows an encoder-decoder design connected by skip pathways that help preserve spatial detail, and has proven effective for medical image segmentation. U-Net++ [17] extends this design with nested, dense skip pathways that narrow the semantic gap between encoder and decoder stages, improving delineation of small or irregularly shaped structures. TransUNet [4] augments a CNN encoder with a Transformer block to capture long-range dependencies alongside local detail, at the cost of additional compute and memory.

2.2 Neuro-Fuzzy Inference Module

The proposed neuro-fuzzy inference module extends the zero-order Takagi–Sugeno (TSK) fuzzy system [11] by integrating learnable Gaussian membership functions and residual feature fusion into a deep neural network, following the differentiable learning principle introduced by ANFIS [5]. Given the bottleneck feature vector $\mathbf{z} = [z_1, z_2, \ldots, z_D]^T \in \mathbb{R}^D$, the module employs K fuzzy rules to infer a refined feature representation. For the j-th fuzzy rule, the membership degree of feature channel d is modeled by a Gaussian membership function

$$\mu_{j,d}(z_d) = \exp\left(-\frac{(z_d - m_{j,d})^2}{2(s_{j,d} + \delta)^2}\right), \tag{1}$$

where $m_{j,d}$ and $s_{j,d}$ denote the learnable center and spread, respectively, and δ is a small positive constant preventing numerical instability. The firing strength of each fuzzy rule is computed by combining all channel memberships and then normalized across the entire rule base,

$$a_j = \prod_{d=1}^{D} \mu_{j,d}(z_d), \qquad \bar{a}_j = \frac{a_j}{\sum_{l=1}^{K} a_l + \delta}, \tag{2}$$

where $\sum_{j=1}^{K} \bar{a}_j = 1$. Unlike the conventional zero-order TSK model that assigns a scalar consequent to each rule, the proposed method associates every rule with a learnable feature vector $\mathbf{v}_j \in \mathbb{R}^D$. The fuzzy inference output is therefore obtained as

$$\tilde{\mathbf{z}} = \sum_{j=1}^{K} \bar{a}_j \mathbf{v}_j, \tag{3}$$

which degenerates to the classical zero-order TSK formulation when $D = 1$.

To preserve the discriminative representation learned by the backbone while allowing adaptive fuzzy refinement, a residual gating mechanism is introduced,

$$\boldsymbol{\gamma} = \sigma\left(\mathbf{U}\tilde{\mathbf{z}} + \mathbf{b}_{\gamma}\right), \qquad \hat{\mathbf{z}} = \mathbf{z} + \boldsymbol{\gamma} \odot \tilde{\mathbf{z}}, \tag{4}$$

where $\sigma(\cdot)$ denotes the sigmoid activation function, $\odot$ is the Hadamard product, and $\mathbf{U}$ and $\mathbf{b}_{\gamma}$ are learnable parameters. Consequently, the fuzzy module can gradually inject rule-based semantic information while preserving the original deep representation through the residual pathway.

All neuro-fuzzy parameters,

$$\Theta = \{m_{j,d}, s_{j,d}, \mathbf{v}_j, \mathbf{U}, \mathbf{b}_{\gamma}\},$$

are optimized jointly with the backbone network using end-to-end back-propagation.

2.3 Training Objective

The proposed network is optimized using a hybrid objective consisting of the Tversky loss, binary cross-entropy (BCE) loss, and two neuro-fuzzy regularization terms. The Tversky index is defined as

$$TI = \frac{\sum_n \hat{y}_n y_n}{\sum_n \hat{y}_n y_n + \alpha \sum_n (1 - y_n)\hat{y}_n + \beta \sum_n y_n (1 - \hat{y}_n)}, \tag{5}$$

where $\hat{y}_n$ and y_n denote the predicted probability and ground-truth label of pixel n, respectively. The parameters α and β control the penalties assigned to false positives and false negatives, with $\beta > \alpha$ adopted to emphasize lesion sensitivity.

The overall optimization objective is

$$\mathcal{L} = \lambda_1(1 - TI) + \lambda_2 \mathcal{L}_{BCE} + \lambda_3 \mathcal{R}_{sharp} + \lambda_4 \mathcal{R}_{sep}, \tag{6}$$

where $\mathcal{L}_{BCE}$ is the binary cross-entropy loss, while the two neuro-fuzzy regularization terms are

$$\begin{aligned} \mathcal{R}_{sharp} &= -\frac{1}{|\Omega|} \sum_{(h,w)\in\Omega} \sum_{j=1}^{K} \bar{a}_j \log(\bar{a}_j + \delta), \\ \mathcal{R}_{sep} &= \frac{2}{K(K-1)} \sum_{j<l} \exp\left(-\frac{\|m_j - m_l\|_2^2}{2\rho^2}\right). \end{aligned} \tag{7}$$

Here, $\mathcal{R}_{sharp}$ encourages confident fuzzy-rule activation, whereas $\mathcal{R}_{sep}$ promotes diversity among fuzzy rules by preventing their Gaussian centers from collapsing.

2.4 Evaluation Metrics

To evaluate image segmentation performance, the Intersection over Union (IoU) and Dice Similarity Coefficient (Dice) are commonly employed. IoU measures the proportion of overlap between the predicted segmentation and the ground-truth region relative to their combined area, thereby reflecting the overall agreement between the two regions. Dice places greater emphasis on the overlapping pixels by considering twice the intersection in relation to the total number of pixels in both regions. Higher values of both metrics indicate better segmentation accuracy, with a value of 1 representing a perfect match between the prediction and the reference annotation. Let M_p and M_g denote the predicted and ground-truth tumor regions. Regional overlap is quantified by:

$$\text{IoU} = \frac{|M_p \cap M_g|}{|M_p \cup M_g|} \quad (8); \quad \text{Dice} = \frac{2|M_p \cap M_g|}{|M_p| + |M_g|} \quad (9)$$

3 Proposed Method

NFTransUNet couples deep encoder-decoder representations with the rule-based reasoning of the fuzzy inference layer of Sect. 2.2, refining the encoder bottleneck through trainable fuzzy rules before decoding so the network can express graded, rule-based confidence about ambiguous tumor boundaries rather than a single hard feature value. The overall architecture is shown in Fig. 1.

3.1 Architecture and Training Pipeline

Training Phase. Data from HDFS is split into train, validation and test sets; the training set is cropped, enhanced, and augmented (rotation, zooming, flipping). The processed images are fed to the feature-extraction block, evaluated with three backbone architectures (U-Net, U-Net++, TransUNet) trained and compared separately. At the bottleneck, the deepest features pass through the neuro-fuzzy layer of Sect. 2.2 before decoding.
Testing Phase. The test data undergoes the same preprocessing. The image is passed through the feature extraction step using the trained model. The system outputs the segmentation result as the original image, the binary mask, and a Grad-CAM [10] overlay highlighting the affected area.

3.2 Applying the Neuro-Fuzzy Model in a Big-Data Environment

Algorithm 2 orchestrates end-to-end training under Apache Spark, wrapping the neuro-fuzzy bottleneck of Algorithm 1 around each backbone and optimizing it with Adam under the settings listed in Table 1. Beyond the pixel-level metrics of Sect. 2.4, we attach Grad-CAM to expose the rule-based feature space, yielding interpretability at both the input and the fuzzy-inference level.

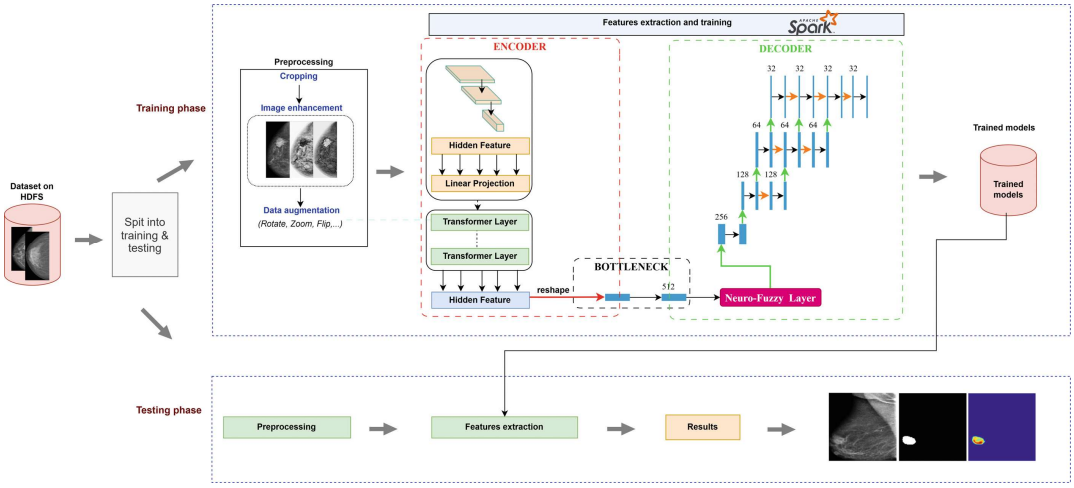


Fig. 1. The proposed NFTransUNet pipeline for tumor segmentation on mammograms in a big-data environment.

Algorithm 1. Forward propagation of the proposed neuro-fuzzy inference layer

Input: Bottleneck feature map Z; learnable neuro-fuzzy parameters Θ; number of fuzzy rules K.

Output: Refined feature map $\hat{Z}$.

for each spatial location (h, w) **do**
  Extract bottleneck feature vector $\mathbf{z} \leftarrow Z(h, w, :)$.
  Compute Gaussian membership values of all fuzzy rules according to Eq. (1).
  Compute and normalize the rule firing strengths using Eq. (2).
  Perform neuro-fuzzy inference to obtain $\tilde{\mathbf{z}}$ according to Eq. (3).
  Compute the adaptive gate and residual fusion using Eq. (4).
  Store $\hat{Z}(h, w, :) \leftarrow \hat{\mathbf{z}}$.
end for
return $\hat{Z}$

Table 1. Experimental scenarios and training parameters

Sce.	Name	Method	Epoch	LR	Batch	Rules
1	U-Net	baseline	150	0.0001	8	–
2		Neuro-fuzzy	150	0.0001	8	4
3	U-Net++	baseline	150	0.0001	8	–
4		Neuro-fuzzy	150	0.0001	8	4
5	TransUNet	baseline	150	0.0001	8	–
6	NFTransUNet	Proposed method	150	0.0001	8	4

Algorithm 2. Distributed training of the proposed NFTransUNet model

Input: Dataset $\mathcal{D}$ stored in HDFS; backbone set B; Spark cluster with P workers.
Output: Trained models, segmentation results, and Grad-CAM visualizations.

```
Split D into training, validation, and testing sets.
Initialize the Apache Spark cluster and distribute the dataset across worker nodes.
for all backbone architecture b ∈ B do
    Construct the baseline and NFTransUNet models.
    for all training mini-batches in parallel do
        Extract bottleneck features.
        Refine bottleneck features using the proposed neuro-fuzzy inference layer (Alg. 1).
        Compute the hybrid loss according to Eq. (6).
        Update local model parameters by back-propagation.
    end for
    Synchronize model parameters across Spark workers.
    Save the trained model to HDFS.
end for
Evaluate the trained models on the testing set.
Generate Grad-CAM visualizations for qualitative analysis.
```

4 Experimental Results

4.1 Environment and Datasets

The single-node setup uses an Intel Xeon Gold 6242 CPU (4 cores, 6GB RAM); the distributed setup uses one master node (32GB RAM) and twelve worker nodes (4 cores, 6GB RAM), running Ubuntu 20.04, OpenJDK 8, Hadoop, and Spark. VinDr-Mammo [7] is a public mammography dataset; we use a curated subset of 4,428 images with expert annotations, comprising 1,696 images with benign masses, 720 images with malignant masses, and 2,012 normal images. VL-Bre is an in-house dataset of 4,360 mammograms collected from a regional oncology hospital and annotated by specialists, comprising 1,108 images with benign masses, 1,260 images with malignant masses, and 1,992 normal images. Both datasets are split 70/15/15 for training, validation, and testing.

4.2 Training and Testing Results

Table 2 shows that the fuzzy bottleneck consistently improves segmentation across both datasets: enhanced models gain approximately 1.1–1.3 Dice points on the test sets, with corresponding IoU gains, indicating more accurate lesion delineation. NFTransUNet (Sce. 6) delivers the best performance (93.50% on VL-Bre, 93.00% on VinDr) with a small train-test gap, suggesting good generalization.

To ensure that the observed improvements of the neuro-fuzzy variants are mathematically robust and not artifacts of a fortunate initialization, we retrained all baselines (Sce. 1, 3, 5) and their corresponding neuro-fuzzy counterparts (Sce. 2, 4, 6) across 5 independent runs using different random seeds. Table 3

Table 2. Segmentation results on VL-Bre and VinDr-Mammo (%).

Sce.	VL-Bre				VinDr			
	Train		Test		Train		Test	
	IoU	Dice	IoU	Dice	IoU	Dice	IoU	Dice
1	89.09	94.23	86.00	91.50	91.46	95.54	84.90	91.00
2	92.40	96.03	87.70	92.80	91.07	95.33	86.60	92.30
3	90.62	95.08	86.60	92.00	91.77	95.71	85.50	91.50
4	90.38	94.95	88.20	93.10	90.80	95.18	87.10	92.60
5	90.31	94.91	87.10	92.40	89.18	94.28	86.00	91.90
6	**91.10**	**95.35**	**88.90**	**93.50**	90.39	94.95	**87.60**	**93.00**

reports the mean Dice scores along with their standard deviations (± STD). The neuro-fuzzy integration consistently yields statistically significant gains; for instance, NFTransUNet (Sce. 6) outperforms the baseline TransUNet (Sce. 5) with a p-value < 0.01 on both datasets. The narrow standard deviations ($\leq$ 0.42%) further verify the stability and reproducibility of our fuzzy-bottleneck design under varying initializations.

Table 3. Statistical significance of the neuro-fuzzy improvements on the test set, mean Dice ± std over 5 independent runs, with paired t-test p-values against the corresponding baseline.

Sce.	Model architecture	VL-Bre Dice (%)	VinDr Dice (%)	paired t-test
1	U-Net (baseline)	91.50 ± 0.35	91.00 ± 0.42	–
2	U-Net (neuro-fuzzy)	$\mathbf{92.80 \pm 0.28}$	$\mathbf{92.30 \pm 0.31}$	< 0.05
3	U-Net++ (baseline)	92.00 ± 0.34	91.50 ± 0.39	–
4	U-Net++ (neuro-fuzzy)	$\mathbf{93.10 \pm 0.27}$	$\mathbf{92.60 \pm 0.29}$	< 0.05
5	TransUNet (baseline)	92.40 ± 0.29	91.90 ± 0.38	–
6	NFTransUNet (ours)	$\mathbf{93.50 \pm 0.22}$	$\mathbf{93.00 \pm 0.25}$	< 0.01

4.3 Distributed Training and Scalability Analysis

Table 4 compares the training performance of the proposed framework on a single node and a 12-node Apache Spark cluster. Across all experimental scenarios, distributed training reduces the overall training time by approximately 6.7×-8.7×, demonstrating good scalability for large-scale mammography datasets. Although a slight reduction in segmentation accuracy is observed when moving from one node to 12 nodes, the decrease in Dice score remains limited (0.1–4.0% points, with an average reduction of approximately 1.8 points). Even under distributed

Table 4. Test results in local (1-node) and distributed (12-node) environments.

Sce.	VL-Bre				VinDr			
	1 node		12 node		1 node		12 node	
	Dice	Time (s)	Dice	Time (s)	Dice	Time (s)	Dice	Time (s)
1	91.50	11600	90.50	1611	91.00	11850	89.00	1723
2	92.80	12700	89.80	1814	92.30	12950	91.30	1924
3	92.00	23470	90.05	2819	91.50	23850	90.00	3109
4	93.10	24620	93.00	3037	92.60	25120	91.60	3238
5	92.40	25210	90.23	2900	91.90	26530	87.90	3262
6	**93.50**	26300	91.00	3190	**93.00**	27600	91.15	3356

execution, all models maintain Dice scores between 87.9% and 93.0%, indicating that the proposed framework preserves segmentation performance while substantially shortening the training time. To further evaluate the scalability of the distributed implementation, we summarize the speedup, parallel efficiency, and the estimated serial fraction for all experimental scenarios on the 12-node Spark cluster. These metrics are computed directly from the training times reported in Table 4. Table 5 reports the speedup $S(N)=T_1/T_N$, efficiency $E(N)=S(N)/N$, and the serial fraction f_s implied by $S(N) \leq 1/(f_s + (1 - f_s)/N)$ for f_s at the observed $S(N)$, N=12, using the timings already in Table 4. All six scenarios reach efficiencies of 0.56–0.72 ($f_s \approx$ 3.5-7.1%), indicating that the workload parallelizes well and that the neuro-fuzzy layer (Sce. 2, 4, 6) adds no disproportionate synchronization cost relative to its baseline.

Table 5. Speedup $S(N)$, parallel efficiency $E(N)$, and serial fraction f_s, computed from Table 4 at $N = 12$ nodes.

Sce.	VL-Bre			VinDr		
	$S(N)$	$E(N)$	f_s (%)	$S(N)$	$E(N)$	f_s (%)
1	7.20	0.60	6.1	6.88	0.57	6.8
2	7.00	0.58	6.5	6.73	0.56	7.1
3	8.33	0.69	4.0	7.67	0.64	5.1
4	8.11	0.68	4.4	7.76	0.65	5.0
5	8.69	0.72	3.5	8.13	0.68	4.3
6	8.24	0.69	4.1	8.22	0.69	4.2

4.4 Assessment on the Number of Fuzzy Rules

Table 6 shows that with ($K = 2$) the fuzzy inference block lacks the capacity to represent diverse tumor appearances, while with too many rules ($K = 6$, $K = 8$)

Table 6. Assessment on the number of fuzzy rules K for NFTransUNet (test set, %).

K	VL-Bre		VinDr	
	IoU	Dice	IoU	Dice
2	85.30	90.40	84.10	89.90
4	**88.90**	**93.50**	**87.60**	**93.00**
6	87.25	90.50	86.90	92.13
8	86.90	91.60	85.80	91.10

the rule bank gains capacity but individual rules specialize less sharply, per the separation term $\mathcal{R}_{sep}$. $K = 4$ gives the best accuracy–complexity balance on both datasets and is used throughout Sect. 4; the statistical significance of the resulting Dice gains is established in Table 3.

4.5 Qualitative Analysis and Interpretability

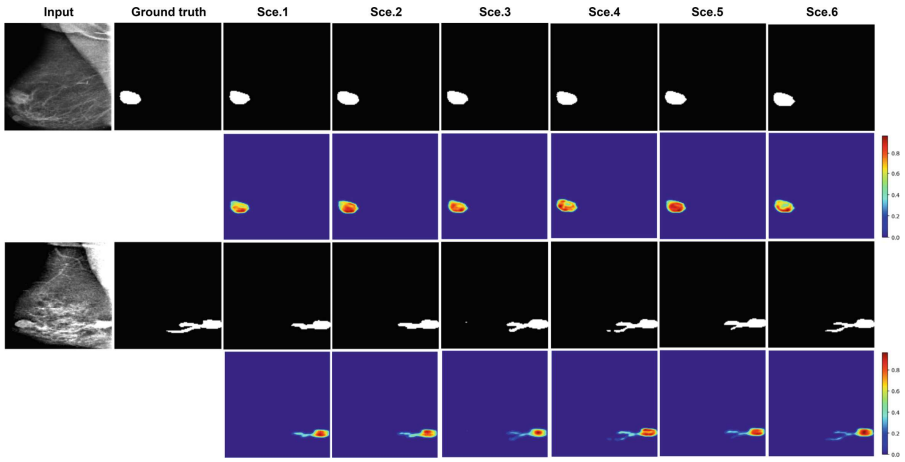


Fig. 2. Segmentation and Grad-CAM results across all scenarios.

Figure 2 shows the predicted masks and Grad-CAM activations for all six scenarios on two representative mammograms. Across both cases, the fuzzy-bottleneck variants (Sce. 2, 4, 6) produce masks whose boundaries more closely follow the irregular contour of the ground truth than their baseline counterparts (Sce. 1, 3, 5), and the Grad-CAM activations of NFTransUNet (Sce. 6) concentrate more tightly on the lesion area, which is qualitatively consistent with the quantitative gains reported in Table 2.

Table 7 enumerates four example rules derived from the trained neuro-fuzzy layer of NFTransUNet, each qualitatively characterized (Low/Medium/High) by the learned Gaussian centers of a selection of bottleneck channels. Rules 1 and

Table 7. Sample fuzzy rules learned by NFTransUNet (bottleneck channels F_{18}–F_{486}; $\|v_j\|_2$ is the L2-norm of the rule's consequent vector)

Rule	F18	F146	F159	F168	F185	F189	F192	F204	F312	F439	F450	F486	$\|v_j\|_2$
Rule 1	Medium	High	High	Low	High	High	Low	Medium	High	High	Low	Low	1.4269
Rule 2	Low	Low	Low	High	Low	High	High	High	High	High	Medium	Medium	1.3963
Rule 3	High	Medium	High	Low	High	Low	Low	Low	High	Low	Low	High	1.4464
Rule 4	High	High	High	High	Low	Medium	High	Medium	Low	Medium	High	Low	1.3977

3 predominantly assign *High* membership to channels such as F_{159} and F_{185}, whereas Rule 2 assigns *Low* membership to these same channels. This indicates that the neuro-fuzzy layer partitions the bottleneck feature space into complementary regions rather than converging on a single dominant rule. The norms of the consequent vectors $\|v_j\|_2$ remain close across rules (1.396–1.446), suggesting that the four rules contribute comparably in magnitude to the combined output.

5 Conclusion

This paper investigates a neuro-fuzzy layer at the encoder bottleneck of a deep segmentation network for tumor segmentation in mammography images, implements a fuzzy inference system with learnable Gaussian membership functions and rule weights, uniformly mounted on three underlying backbone architectures. Fuzzy-enhanced models improve segmentation efficiency, particularly on tumor-containing regions, with best performance achieved by NFTransUNet on both experimental datasets (93.5% Dice on VL-Bre, 93% on VinDr). Comparing local and distributed environments, training time is significantly reduced with more computational nodes (approximately 6.7–8.7 times at 12 nodes); Segmentation accuracy decreased by 0.1-4.0 percentage points (average around 1.8), however, the decrease is acceptable as Grad-CAM visualization showed improved focus on tumor regions, including small lesions. Further work includes testing on multiple datasets and extending the model to other lesion types such as detecting and segmenting microcalcifications.

Disclosure of Interests. The authors have no competing interests to declare.

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